

Evaluating LLMs for Automatic Correction and Detection of Ontology Pitfalls

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Abstract

The construction of ontologies is one of the most relevant tasks in knowledge engineering. The rise of LLMs has recently led the community to introduce changes in knowledge modelling approaches, while little has been done on LLM-assisted evaluation or correction. In this article, we assess the ability of LLMs to identify axioms that solve some given modelling problems, and their ability to identify issues themselves. We adopt the OOPS! metrics as a reference for conceptual modelling errors. The experiment includes ontologies from similar domains: biology, environment, and exposure, in order to limit possible biases due to different vocabularies. This work is part of the interdisciplinary DAE (Détection d’Anomalies Environnementales) project.

Keywords

Ontology, LLM, Evaluation

1. Introduction

Ontology engineering remains a major scientific challenge in the fields of modelling and artificial intelligence [1]. Despite its long history, it continues to stimulate the community thanks to new technologies and needs that regularly emerge. An “ontology” constitutes a formal representation of the elements that make up a specific domain [2]. It is useful to conceive of these domain elements as nodes (independent concepts) and edges (connections between concepts). We can then define a “knowledge graph” as a data graph designed to accumulate and convey knowledge about the real world, where the nodes represent entities of interest and the edges represent the relationships between these entities [3]. In some contexts, the two terms are used interchangeably. Thanks to the expressiveness of the formal languages used to describe ontologies, it is possible to introduce logical reasoning to infer new knowledge or detect inconsistencies in existing information. Because of its human-designed nature, much effort has been devoted to determining how to evaluate the results of the ontology construction process. Firstly, the quality of an ontology is defined by its ability to be shared or used by those seeking information on the domain it represents. Furthermore, advances in machine learning have led to a suite of algorithms for estimating the added value of a knowledge graph.

“Dynamic” evaluation reflects an ontology’s ability to improve performance on specific tasks. In other words, the added semantics enables improvements in, for example, the prediction of missing links or classification. In contrast, “static” evaluation takes into account the way in which the model has been formalised. Although modelling helps with certain objectives, errors are generally present. Even today, the best-known work on construction errors is *Ontology Pitfall Scanner! (OOPS!)* [4], in which a list of conceptualisation pitfalls has been published. The list includes over forty common errors; we report in Table 1 the list of the pitfalls detected by the online tool¹.

Thanks to the enormous capacity of *Large Language Models (LLM)* [5] to understand use cases in

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¹<https://oops.linkeddata.es/>

Code	Meaning
P02	Creating synonyms as classes
P03	Creating the relationship "is" instead of using "rdfs:subclassOf", "rdf:type" or "owl:sameAs"
P04	Creating unconnected ontology elements
P05	Defining wrong inverse relationships
P06	Including cycles in a class hierarchy
P07	Merging different concepts in the same class
P08	Missing annotations
P10	Missing domain information
P11	Missing disjointness
P12	Missing domain or range in properties
P13	Equivalent properties not explicitly declared
P19	Defining multiple domains or ranges in properties
P20	Misusing ontology annotations
P21	Using a miscellaneous class
P22	Using different naming conventions in the ontology
P24	Using recursive definitions
P25	Defining a relationship as inverse to itself
P26	Defining inverse relationships for a symmetric one
P27	Defining wrong equivalent properties
P28	Defining wrong symmetric relationships
P29	Defining wrong transitive relationships
P31	Defining wrong equivalent classes
P34	Untyped class

Table 1
OOPS! pitfalls for every ontology.

written form, they can be used by experts to partially automate certain stages of construction. Work following this line of inquiry has shown that the introduction is generally favourable, whilst a defined approach incorporating them does not currently exist. In fact, the absence of a shared methodology threatens the reproducibility and reliability of the results.

As part of the DAE project, we have developed the OntoPFAS² ontology, incorporating various static analyses, including OOPS! One of the objectives, therefore, is also to examine the capacity of LLMs to improve our modelling. OntoPFAS is an ontology dedicated to representing PFAS molecules [6] in the environment. It is a resource that brings together several fields of interest, such as chemistry, geography, and exposure to substances and their effects. Hence, we have collected ontologies recognised within the community on related topics. Several language models were tested on five ontologies that deal with broadly similar concepts or are interrelated.

The remainder of this article is organised as follows: Section 2 presents the state of the art regarding methods and use cases for LLMs in the field of ontology construction. Section 3 introduces the ontologies used in our experiment. Section 4 describes our evaluation approach, as well as the necessary pre-processing steps. Section 5 presents the results of the LLMs across several improvement tasks and the general insights supported by the results. Section 6 discusses the statistical significance of the results. Finally, Section 7 summarises the results, explores potential extensions of this work, and concludes the article.

2. Related Work

Methods for the static evaluation of ontologies The static evaluation of ontologies allows us to verify best practices in the modelling process. Typically, several aspects are taken into account,

²<https://github.com/davide1797/OntoPFAS>

including the presence of metadata, the absence of isolated elements, the comprehensiveness of concepts, and a consistent naming convention. OOPS! ³ is a service (also available online) that is able to detect the majority of modelling errors cited in the original paper. Among the *pitfalls*, we list here the most frequent ones based on our experience: *Creating unconnected ontology elements*, *missing annotations*, *missing domain or range in properties*, *inverse relationships not explicitly declared*, *using different naming conventions in the ontology*.

Although OOPS! metrics are the best known and almost mandatory for static ontology analysis, there are also other evaluations. Alongside OOPS!, *FAIR OOPS (FOOPS)* [7] identifies missing metadata that impacts *FAIR (Findable, Accessible, Interoperable, Reusable)* aspects. These properties are increasingly expected in newly released software. FOOPS is also an online tool⁴ that checks for the presence and consistency of triples that ensure the resource can be reused correctly. To illustrate, the following properties correspond to the respective aspects: *Ontology has a persistent URL* (findable), *Ontology uses an open protocol* (accessible), *Ontology imports or reuses well-established vocabularies* (interoperable), *Ontology documentation: all terms have labels* (reusable).

Other evaluations are also possible. Vrandečić et al. [8] proposed characteristics that define the quality of an ontology according to several aspects – with a strong overlap with FAIR – but there is no method for the automatic detection of these defects. Furthermore, there are methods requiring human interaction, such as validation via SPARQL queries [9] of *competence questions* [10] or through interfaces and use cases [11, 12].

LLMs in ontology construction The ability of LLMs to adapt to different domains, vocabularies and requirements has led knowledge engineers to rely on them and to partially incorporate them into ontology construction or evaluation tasks. A very common use case involves the verification of ontology axioms, as proposed by Tsaneva et al. [13]. Axioms can be translated into natural language and provided as input for the *prompt*. Rebboud et al. [14] evaluated large language models against the main tasks of ontology construction: conceptualisation, formalisation of query conditions, writing SPARQL queries, and generating documentation. However, the work does not focus on validation, but rather on query writing. Lippolis et al. [15] evaluated the main language models in detail, primarily in a semi-automated manner. The automated quality assessment also focused on writing SPARQL queries, without considering the possibility of correcting axioms. Finally, work such as [16] demonstrates the possibility of training LLMs through *fine-tuning* to meet specific requirements in certain domains.

Overall, LLMs are increasingly being incorporated into the ontology construction process. At present, a shared methodology does not yet exist, but initial attempts are promising. Furthermore, state-of-the-art evaluations focus on query writing and still neglect the static aspects highlighted by OOPS! Therefore, this work can serve as a first step towards assessing and understanding the extent to which LLMs are capable of introducing relevant axioms whilst preserving the semantics of what they represent. The specific nature of the OOPS! pitfalls ensures an accurate assessment of the types of changes they predict.

3. Data Sources

In order to compare the results of our ontology, we have taken into account resources in similar fields. Here we provide a detailed description of our ontology, as well as brief descriptions of other well-known knowledge bases.

OntoPFAS [17] is a conceptualisation of the PFAS domain, covering its aspects of exposure, measurement, and propagation. Combining chemical, geographical, and sensor data, it aims to serve as a platform for practitioners from various fields wishing to contribute data, models or artificial intelligence applications. With the parallel development of the PFAS Data Hub dataset⁵, we have laid the foundations

³<https://oops.linkeddata.es/index.jsp>

⁴https://foops.linkeddata.es/FAIR_validator.html

⁵<https://pdh.cnrs.fr/en/datasets/>

for a positive synergy between data availability and the solutions on offer. Many concepts are drawn from existing ontologies: *ChEBI*, *ExO*, *ENVO* and *SOSA*, *Schema.org*, *PROV-O* and *DCAT*.

This ontology primarily represents three aspects of PFAS-related concerns: chemistry, measurements, and exposure. The graphical architecture provides an overview of the ontology. It is available on GitHub⁶ as well as on *AgroPortal*⁷.

Exposure Ontology (ExO) [18] aims to bridge the gap between exposure science and environmental health disciplines by facilitating the centralisation and integration of exposure data to inform our understanding of environmental health.

Green-AI Ontology [19] is the first attempt to model the energy consumption of artificial intelligence models. The knowledge base includes metrics and tools for measuring consumption across various aspects. The reuse of existing concepts is encouraged by the wealth of concepts already available in the fields of energy, the environment, and computer science.

Agri-Food Experiment Ontology (AFEO) [20] is an ontology network developed on the basis of two existing resources: the Ontology for Agriculture Experiment (OAE) and the Ontology for Food Processing Experiment (OFPE). It contains 136 concepts covering various viticultural practices, as well as products and operations related to wine production.

~OntoPFAS is a deliberately altered version of the OntoPFAS ontology in which all the implemented pitfalls⁸ in OOPS! are present. To provide better support, we introduced at least 5 items for every pitfall not present in the other ontologies. There are binary pitfalls like the absence of a declared licence for which a cardinality does not apply, and for which only the presence (✓) is indicated. They, however, represent a minority. The objective is twofold: (i) to understand how the performance of LLMs varies according to the type of error, and (ii) to understand whether the number of pitfalls impacts performance or leads to inconsistent corrections.

4. Approach

In this section, we describe the various steps involved in comparing LLMs with regard to ontology evaluation. We performed two tasks, a *supervised* and an *unsupervised* one. The first one, based on OOPS!, aims to recognize the ability of the models in fixing the OOPS! pitfalls. It is supervised since the report of the present pitfalls is provided by the OOPS! API. The latter is unsupervised since we ask the models to automatically understand possible pitfalls and introduce axioms to fix them.

1. OOPS! inspection Before querying the LLMs, we use the OOPS! service as an initial source to understand which are the possible improvements. The result of the analysis will form part of the prompt to be provided to the LLMs. The analysis contains a list of errors with the associated concepts, conceived as *ground truth*. We present in Table 2 the list of identified pitfalls for every ontology.

2. Pre-processing For this experiment, we tested several open-access language models. As they are limited in terms of prompt size, we adapt the knowledge bases and the OOPS! results so that they can be processed, whilst retaining the information necessary for correcting the axioms. Taking into account all possible OOPS! pitfalls, we have selected a subset of predicates that enables LLMs to understand the meaning of concepts, as well as the semantic information of the predicates and the hierarchies. Finally, we propose a minimal list of predicates that are sufficient to correct the pitfalls: `rdf:type`, `rdfs:label`, `skos:definition`, `owl:inverseOf`, `rdfs:range`, `rdfs:domain`,

⁶<https://github.com/davide1797/OntoPFAS>

⁷<https://agroportal.lirmm.fr/ontologies/ONTOPFAS>

⁸<https://oops.linkeddata.es/webservice.html>

Pitfall	OntoPFAS	ExO	Green AI	Agri-Food	~OntoPFAS	Support
P02					7	7
P03					✓	1
P04	1	32	1	8	16	58
P05					9	9
P06					5	5
P07					5	5
P08		146	2	76	57	281
P10			✓	✓	✓	3
P11	20	8		8	25	61
P12					✓	1
P13		8	18	8	12	46
P19					5	5
P20					✓	1
P21				✓	✓	2
P22	✓	✓		✓	✓	4
P24					5	5
P25					5	5
P26					6	6
P27					6	6
P28					5	5
P29					6	6
P31					✓	1
P34			2		5	7
Tot	22	195	24	103	186	-

Table 2

OOPS! pitfalls for every ontology.

Ontology	# axioms before	# axioms after
OntoPFAS	1 616	616
ExO	2 355	493
Green AI	581	234
Agri-Food	521	197
~OntoPFAS	1 701	650

Table 3

Number of ontology predicates before and after preprocessing

`rdfs:subClassOf`, `rdfs:subPropertyOf`. For the OOPS! results, we are not interested in trying to resolve *every* error, but rather to understand the capability of LLMs in dealing with the pitfall. That is why we fix 5 as the maximum number of involved items for each pitfall. We also saved some space by removing redundant XML tags in the result. We demonstrate, for the ontologies, the number of axioms before and after the preprocessing in Table 3, and, for the OOPS! results, the amount of space saved in Table 4.

3. Prompts After checking that the length of the knowledge bases does not exceed the maximum size of the LLMs under consideration, we enrich our prompt with the detailed response provided by OOPS! and a prompt that is fairly generic but clear about the objectives. We reiterate the two tasks: (i) introducing axioms to resolve OOPS! pitfalls, and (ii) modifying or adding axioms regarding modelling issues that it has itself detected. We propose the following prompts:

Ontology	# kB before	# kB after
OntoPFAS	3.9	0.6
ExO	19.9	1.2
Green AI	4.8	0.8
Agri-Food	15.2	2.2
~OntoPFAS	29.5	5.4

Table 4

Occupied space before and after preprocessing for the OOPS! results

- You are an ontology engineering expert. Based on the ontology and the OOPS! pitfalls in this prompt, if possible, find concrete axioms that resolve them, provided they do not introduce new ontological problems.
- You are an ontology engineering expert. Based on the ontology, identify potential ontological pitfalls and propose solutions, including concrete axioms where possible.

These two prompts were the result of several experiments in which we did not observe any significant improvement when varying the prompt. Ultimately, the phrase “if possible” helps in not forcing the LLM to propose suggestions when it is not really aware of how to do it.

4. Supervised evaluation For the first part of the evaluation, we focus on the impact of axioms on OOPS! pitfalls; in other words, whether adding the axioms eliminates the detected pitfall(s). In fact, we count the pitfall as “resolved” if the suggestion is also realistic within the context. For example, if OOPS! detects that a property lacks a “domain”, it is not enough for the LLM to simply add an axiom for this: it must also make sense within the model. This evaluation is performed by an author of this paper, who was responsible for the creation of one of the tested ontologies and familiar with the represented vocabularies.

5. Unsupervised evaluation For unsupervised evaluation, two metrics are calculated: (i) the ability to identify actual modelling errors and (ii) the ability to resolve them. Overall, two scores provide a measure of their competence on the task.

Among the freely available models⁹, we selected the best-known ones based on ease of reuse. Thus, we based our work on the *OpenRouter*¹⁰ interface, so that the experiments are easily replicable and no model is given an advantage. Furthermore, the common interface ensures that conditions are equalised across all constraints (response time, data size, etc.). Finally, the following models were tested: (M)istral-7b-instruct, (L)AMA-3-8b-instruct, (G)emma-3-4b-it, (D)eepseek-r1t-chimera:free, and (G)oogle (G)emini-embedding-001. All the evaluations, LLM responses, code, and resources are freely available online¹¹.

5. Evaluation

Numerical evaluation For task (i), we demonstrate the ability of LLMs to resolve the various pitfalls that can be found in OOPS! For this evaluation, it was necessary to create a knowledge base such as ~OntoPFAS which includes all errors not present in the others. Although a single occurrence of a few errors is not statistically robust, it gives us a glance of the models’ ability to handle them. We can thus measure, for each model, the ratio of times they propose axioms that are actually resolute and consistent. Table 5 shows the ratios (best results are shown in bold).

⁹Note that policies of prices may change

¹⁰<https://openrouter.ai/>

¹¹https://github.com/davide1797/oops_pitfalls_correction

Pitfall	Mistral	Llama	Gemma	DeepSeek	Gemini
P02	5/5	0/5	0/5	5/5	5/5
P03	0/1	0/1	0/1	1/1	1/1
P04	12/17	9/17	4/17	13/17	13/17
P05	5/5	0/5	0/5	0/5	0/5
P06	5/5	0/5	0/5	5/5	3/5
P07	5/5	0/5	0/5	5/5	5/5
P08	17/17	10/17	9/17	17/17	16/17
P10	1/3	1/3	1/3	2/3	3/3
P11	16/20	16/20	6/20	20/20	20/20
P12	1/1	0/1	0/1	0/1	1/1
P13	15/20	12/20	10/20	20/20	17/20
P19	5/5	0/5	0/5	5/5	5/5
P20	1/1	0/1	0/1	1/1	1/1
P21	1/2	0/2	0/2	0/2	0/2
P22	2/4	1/4	1/4	3/4	3/4
P24	5/5	0/5	0/5	5/5	5/5
P25	5/5	0/5	0/5	5/5	0/5
P26	5/5	0/5	5/5	5/5	5/5
P27	5/5	0/5	0/5	5/5	0/5
P28	5/5	0/5	0/5	5/5	0/5
P29	5/5	0/5	0/5	5/5	5/5
P31	1/1	0/1	0/1	1/1	0/1
P34	7/7	0/5	0/5	7/7	0/7
Accuracy	129/149	49/149	36/149	135/149	109/149

Table 5
Evaluation of models for every pitfall.

Next, we also show the distribution of corrections for knowledge base and pitfalls, respectively in Tables 6 and 7. In these two tables, we limit the overview to present only if the pitfall has been solved at least once or not, to not overwhelm the visualization.

Next, for task (ii), we asked the LLMs to detect anomalies in the modelling, as well as to propose axioms that improve it. For each model and ontology, we count the number of suggested pitfalls and, among these, how many are *plausible* corrections (not hallucinations). Figure 1 shows, using bars, the number of proposed corrections, with ontologies distinguished by colour. In contrast, the number of true corrections is shown by diagonal lines. Table 8 demonstrates the ratios between plausible axioms and the total of axioms.

Discussion Although LLMs rarely produce results that are perfectly explainable, we have identified some general patterns. The first finding that emerges is the difference in providing corrections between ExO and the rest. This difference is not linked to pitfalls, as the same errors are found in the other ontologies. What we have noted is the difference in the clarity and specificity of the concepts. Indeed, there are many modelled concepts that lack clear distinctions, and the descriptive axioms are weak. On the other hand, we observe that Gemini is considerably more robust than the others for this task, although not among the best ones for the first one. By looking at Table 1, it is evident that the introduction of several pitfalls in $\sim\text{OntoPFAS}$ also allows the generation of more suggestions, although surprisingly

Ontology	Pitfall(s)	M	L	G	D	G
OntoPFAS	4				✓	
	11	✓		✓	✓	
	22	✓	✓	✓	✓	
ExO	4, 8	✓			✓	
	11, 13, 22		✓		✓	
Green AI	4, 8, 10, 13, 34	✓	✓	✓	✓	✓
Agri-Food	4, 10, 21, 22					
	8	✓				
	11	✓	✓		✓	✓
	13	✓			✓	
~OntoPFAS	2, 4, 6, 7, 13, 19, 20, 24, 29	✓			✓	✓
	3				✓	✓
	5, 21	✓				
	8, 11, 22, 26	✓		✓	✓	✓
	10	✓	✓	✓	✓	✓
	12	✓				✓
	25, 27, 28	✓			✓	
	31, 34	✓		✓		

Table 6

Evaluation of models for every ontology.

-	02	03	04	05	06	07	08	10	11	12	13	19	20	21	22	24	25	26	27	28	29	31	34
M	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
L			✓				✓	✓	✓		✓				✓								✓
G			✓				✓	✓	✓		✓				✓			✓				✓	✓
D	✓	✓	✓		✓	✓	✓	✓	✓		✓	✓	✓		✓	✓	✓	✓	✓	✓	✓		✓
GG	✓		✓		✓	✓	✓	✓		✓	✓	✓	✓			✓		✓			✓		✓

Table 7

Evaluation of models for every pitfall.

only in two cases the models suggest the greatest number of axioms for it. Generally speaking, the Gemini and DeepSeek models demonstrate greater effectiveness than the others in correction.

Finally, we highlight that the language can be a source of confusion. For example, DeepSeek attempts to resolve pitfall P13 (“Inverse relationships not explicitly declared”) with incorrect relation names; for instance: `exo:#is_associated_with> owl:inverseOf exo:# associated_to>` but “associated to” is not a correct expression in English, and even if it were, the meaning would be a synonym of the original relation. In many cases, in addition, the proposed axioms aim at solving the pitfall quickly rather than providing a more robust solution: for instance choosing **owl:Thing** as domain or range of properties.

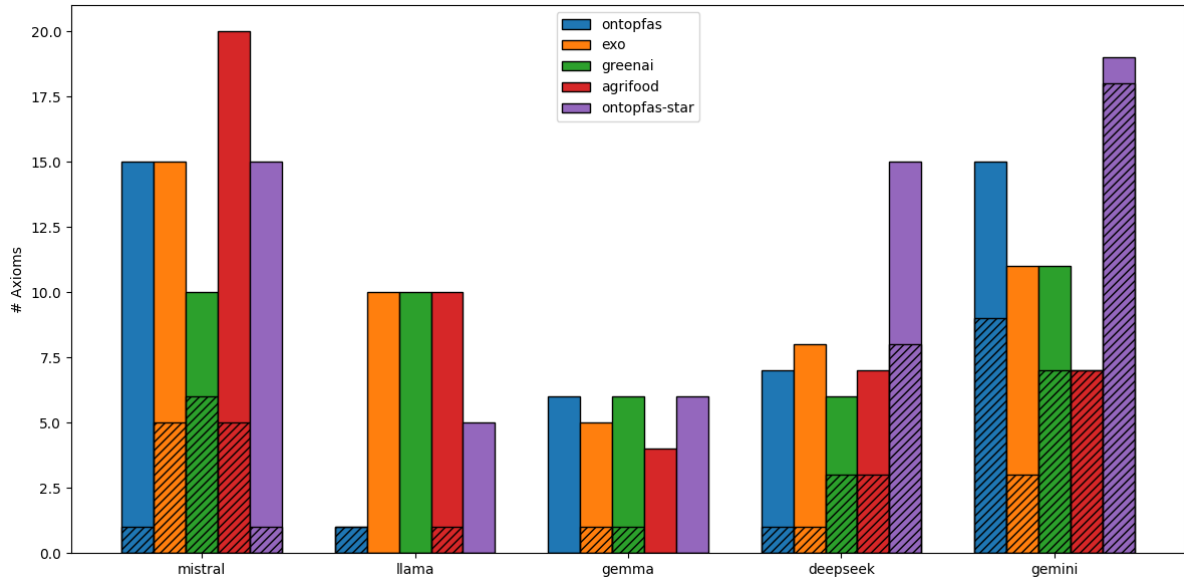


Figure 1: Chart of the added pitfalls per LLM and ontology

Ontology	Model	Accuracy
OntoPFAS	Mistral	0.07
	Llama	1.00
	Gemma	0.00
	DeepSeek	0.14
	Gemini	0.60
ExO	Mistral	0.33
	Llama	0.00
	Gemma	0.20
	DeepSeek	0.13
	Gemini	0.27
Green AI	Mistral	0.60
	Llama	0.00
	Gemma	0.17
	DeepSeek	0.50
	Gemini	0.64
Agri-Food	Mistral	0.25
	Llama	0.10
	Gemma	0.00
	DeepSeek	0.43
	Gemini	0.64
~OntoPFAS	Mistral	0.07
	Llama	0.00
	Gemma	0.00
	DeepSeek	0.53
	Gemini	0.95

Table 8
Precision of suggested pitfalls.

6. Statistical Significance

We may be tempted to draw general conclusions on LLMs' performance for this task by looking at the occurrences of resolved pitfalls/axioms. However, given the technical limitations of free LLMs and the OOPS! API, and the lack of a dataset for this task, the impact of the choices of our dataset on statistical

significance has to be assessed. In this section, we provide statistical tests to understand what results are robust enough to be used to depict some conclusions.

Statistical significance will be evaluated for both tasks (i) and (ii), and the analysis will be done, respectively, for:

- *model*, *pitfall*, and *ontology* for task (i),
- *model* and *ontology* for task (ii).

6.1. Pitfall Solving

In this section, we answer three questions starting from the results of Tables 5 and 6:

What are the best models to solve pitfalls? For this part, we do not consider priorities of pitfalls; hence, the overall accuracy is used to understand the significance of the performances. To compare the models, we used the *contingency table* and the χ^2 test to verify the existence of a significance. As expected from the different accuracies, the test outputs a value that is much lower than 0.001, confirming our hypothesis. Less trivial is the pairwise comparison between the different models. With 5 models, we have 10 pairwise comparisons. We provide here the 3 pairwise comparisons that are not statistically significant: Mistral-DeepSeek, Mistral-Gemini, and Llama-Gemma. For the other possible pairs, the difference in overall accuracy seems to better reflect a model’s capacity to address the pitfalls. Trying to build a ranking of models for this task and datasets, we use the symbols $<$ to indicate that a model performs worse than another and $\leq$ to indicate that a model performs worse than another but not with statistical significance. The pairwise comparisons showed the following relationships:

- Gemma $\leq$ Llama $<$ Gemini $\leq$ Mistral $\leq$ DeepSeek
- Gemini $<$ DeepSeek

We also analyze the significance for the *important* pitfalls, those ranked as “important” or “critical” in the OOPS! catalogue¹². In this analysis, there are no surprises in the ranking, but there is more statistically significant distances between the models:

- Gemma $\leq$ Llama $<$ Gemini $<$ Mistral $\leq$ DeepSeek

Are there pitfalls for which changing models make an impact? Among the pitfalls, given the lack of a great support (boolean) or their rare presence, we exclude from the analysis 7 pitfalls that are not significantly meaningful: P03, P10, P12, P20, P21, P22, P31. They can also be considered the most difficult ones, although their cardinality is a limit.

Other pitfalls are highly susceptible to the change of models like P05, P07, P11, P26, and P34, each of them with p-value much lower than 0.001 considering the comparison between models. By considering the ratio between the accuracy for each pitfall and model, P08, P26, P11, and P13 are the easiest to solve.

Which ontologies are the easiest to handle? By referring to Table 6, we consider only if a model is capable of solving at least one instance of the pitfall for every ontology. We check whether some of the chosen ontologies are (statistically) easier than others. The contingency table and the χ^2 analysis on it shows that there is a high correlation, but especially thanks to the Green AI ontology. With the pairwise analysis, we obtained the following result (with $<$ meaning “easier than”):

- Green AI $<$ OntoPFAS $\leq$ ExO $<$ Agri-Food $\leq$ ~OntoPFAS

We attempt to provide an explanation for this result. One of the reason why we introduced the altered version (as previously specified) is not only to test all the pitfalls, but also to verify if increasing the number of pitfalls complicates the resolution of the model. The obtained results seem to confirm our

¹²<https://oops.linkeddata.es/catalogue.jsp>

hypothesis, although there are also other aspects to take into account, like the introduction of several minor (and less known) pitfalls that are likely harder to correct. On the other hand, we also expected that the available annotations would have an effect, but this is not confirmed since `OntoPFAS` and `~OntoPFAS` presents exactly the same metadata information.

6.2. Axioms Suggestion

In this section, by taking into account Figure 1 we answer the following questions:

Do some ontologies induce models to generate more suggestions? To answer this question, we use the Friedman test [21]. The test verifies if there is a significant difference in the average of variables, in our case to understand whether there is an ontology that is more suitable to be “corrected”. The test presents a p-value much higher than the threshold, rejecting the null hypothesis; i.e., in the data, there is no correlation between the ontology and the number of provided suggestions. If this result, which surprises us, is due to the limitation of free LLMs or to how the model interprets the task will be further investigated. What was more expected is that there are LLMs providing more suggested axioms than others.

Do some ontologies induce models to generate more correct suggestions? Referring to the same test, we discovered here that the p-value is lower than the one for the previous question, although still above the threshold.

In conclusion, the unsupervised task seems not susceptible by the provided ontology, while differences between the number of provided (and correct) axioms vary according to the models.

7. Conclusion

In conclusion, we summarise the contribution of this work: the evaluation of LLMs’ ability (i) to resolve OOPS! pitfalls and (ii) to propose and correct new modelling pitfalls. Within the context of free LLMs, we have reduced the size of the knowledge bases mentioned whilst retaining the aspects necessary for understanding the semantics. Finally, the evaluation results show that some models are clearly more capable than others for these tasks and that some ontologies can be easier to fix. This will likely occur when the source contains relatively few pitfalls. This work opens up several avenues for research, notably the improvement of tasks using *fine-tuning* or *prompt engineering* techniques, the large-scale evaluation of commercial models, and the measurement of other quality metrics like FOOPS. We claim that this task can be further investigated and, from this perspective, we are considering working on a creation of a dataset of pitfalls based on larger ontologies.

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Declaration on Generative AI

During the preparation of this work, the authors used Grammarly to check grammar and spelling. The authors take full responsibility for the publication’s content.

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